JC-527299 Statement of Qualifications, Michael Peña

As a research-based Statistical Consultant with a strong background in programming, exploratory data analysis, data processing, and data visualizations, I am highly interested in the Geographic Information System Analyst position under the California Department of Forestry and Fire Protection. I recently contributed my statistical work in my submission for academic literature. Data visualizations are key for review, including peer reviewed publishing, to check data integrity, accuracy, and quality. Through a cooperative team environment within a multi-disciplined team, I helped coordinate development as we extracted raw data from a lab database and processed it through data cleaning, data validation, and exploratory analysis before being visualized using box plots, bar charts, and bivariate plots to understand the story the data was trying to tell us. I conducted systems analysis on the pipeline to ensure efficiency, and tested validation scripts to maintain standards for accuracy. Visualizations were tailored for presenting to our client, where complex information could be effectively communicated through presentations, briefings, and provide reports to stakeholders.

Sophisticated data visualizations of this data are seen by stakeholders. These visualizations were labeled using tools in R, particularly with the ggplot2 package, allowing color and texture to improve readability and data interpretation. I would install custom packages and modify code to automate repetitive processes, looping observational data for different attributes from datasets. Developing these visualizations helped support our research in gathering insight for decision-making in selecting models or statistical inference methodologies related to biomechanical research. I applied creativity and innovation to design visualizations that improved communication of complex statistical results while monitoring quality and progress throughout the project lifecycle.

I formulated hypotheses and implemented a dimension reduction statistical method on numerical raw data for our specific statistical model, coding a visualization to help decide how many dimensions should reduce the numerical data to. I coded this visualization using R for publication readiness, ensuring outputs were print and digital-ready, allowing resizing without image distortion, in which the reduced dimensions were principally composed as aspects of the unreduced data. I built a principal component vector plot to visualize how each variable affected the principal component, which justified the recommended decision to reduce the data. After performing dimension reduction, I focused communication on how the K-nearest neighbors selection model was working for our observed data by creating innovative visualizations with titles, labeled axes, and color-coded legends that mapped the regions depicted from the model. These analyses required me to independently implement statistical methodologies and effectively communicate research findings through written and verbal communication skills.

In the data pipeline process of developing data visualizations, if data outliers or missing data were observed, I analyzed the dataset using data cleaning and data validation practices to assess the effect on data accuracy in our research publication draft as well as reliable datasets. I would prepare datasets for analysis and provide assistance to team members on statistical methodology when needed. This process required me to adapt to changing priorities and quick deadlines while maintaining data quality throughout the research workflow. Hence, my dedication to data quality and breadth of experience in statistical methods, tools, and models qualifies me for this position.